Indicators of Heavy Traffic on I-94

An exploratory data analysis of the Metro Interstate Traffic Volume dataset to identify key indicators — including time of day, day of week, season, and weather — that predict heavy traffic on Interstate 94.

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Dataset Overview

Dataset Snapshot

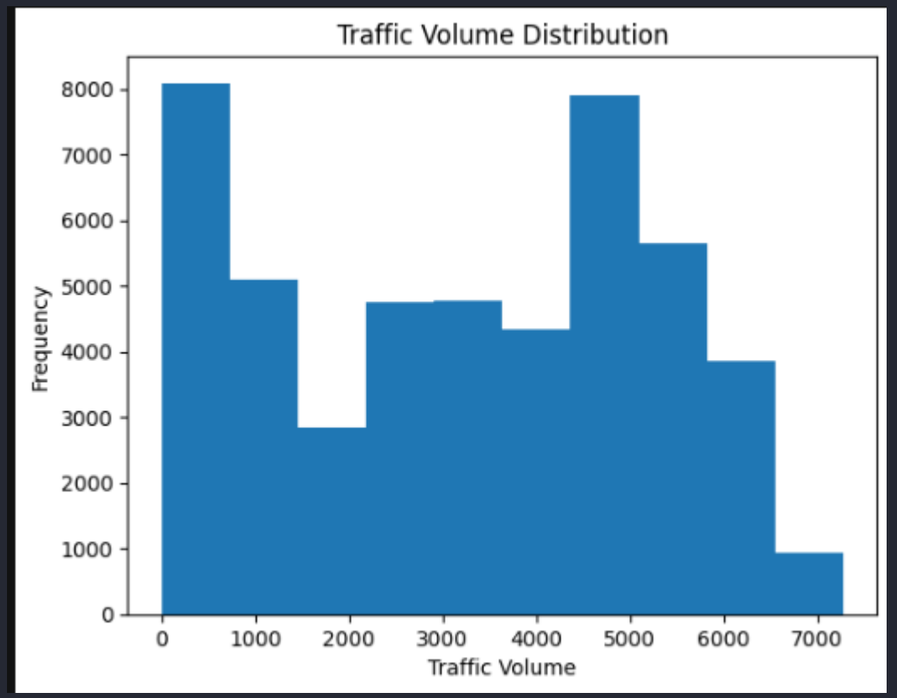
48,204 rows *× 9 columns covering hourly traffic and weather data from* ***2012-10-02*** *to* ***2018-09-30 spanning over 6 years****.*

Columns include: ***holiday, temp, rain_1h, snow_1h, clouds_all, weather_main, weather_description, date_time, traffic_volume****.*

Key Data Quality Notes

- *All columns are fully populated* ***except holiday****, which has only 61 non-null values (recorded at midnight on each holiday)*
- *48,143 holiday entries are null — treated as "No Holiday"*
- *Traffic volume ranges from* ***0 to 7,280*** *cars/hour, with a mean of* ***3,260***
- *Bottom 25%: ≤1,193 cars/hour; Top 25%: ≥4,933 cars/hour*

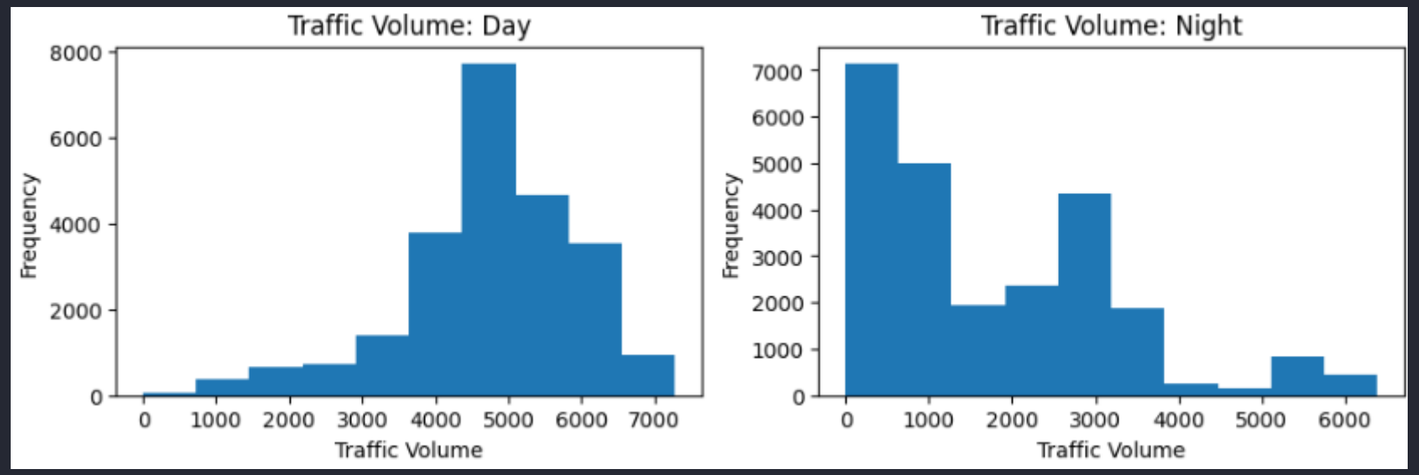
48,204	9	3,260	7,280
Total Records	Columns	Avg Cars/Hour	Max Cars/Hour



Traffic Volume Distribution

*The distribution is **bimodal**: a primary peak at the 0–500 bin (~8,100 occurrences) reflects nighttime/low-activity hours, while a secondary peak at 4,500–5,000 (~7,900 occurrences) reflects busy daytime commuting. This contrast motivates splitting the analysis into **daytime vs. nighttime** data.*

Day vs. Night Traffic



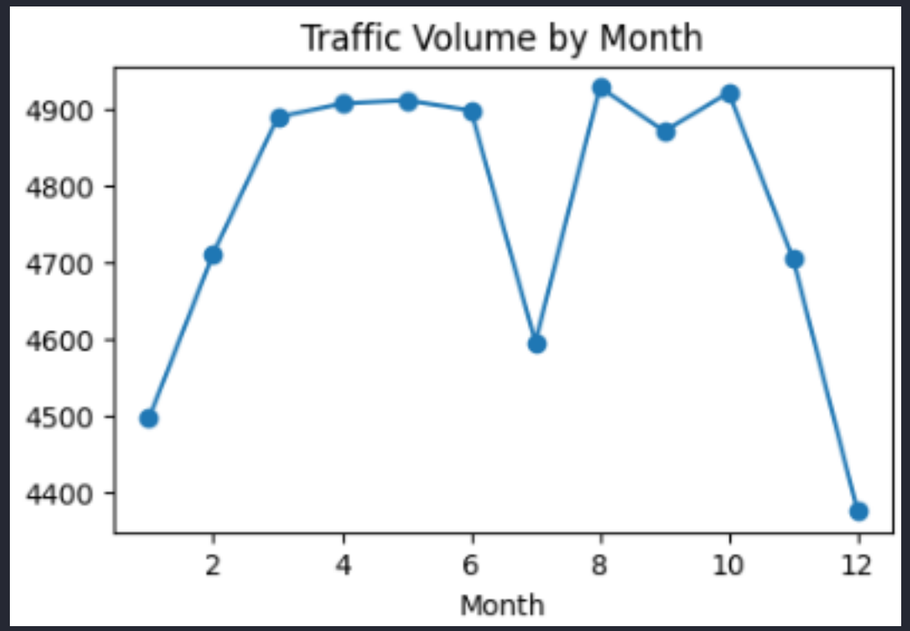
Daytime (7 AM – 7 PM)

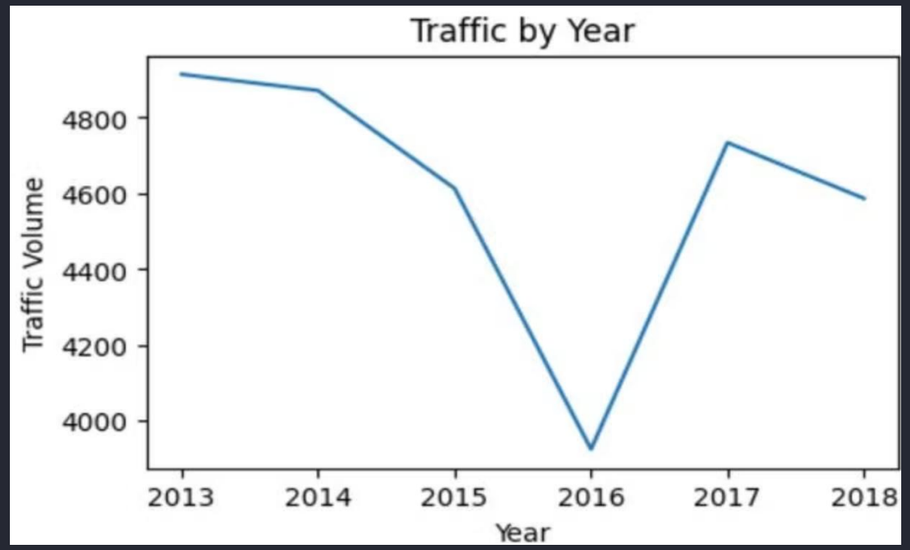
*Left-skewed distribution. Mean: **4,762** cars/hour. 75% of observations exceed **4,252** cars/hour. Traffic is consistently high.*

Nighttime (7 PM – 7 AM)

*Right-skewed distribution. Mean: **1,785** cars/hour. 75% of values fall below **2,819** cars/hour. Traffic is mostly low.*

*Since heavy traffic occurs primarily during the day, all further analysis focuses on **daytime data**.*





TIME INDICATORS

Traffic by Month & Season

*Traffic is heavier during **warm months (March–October)**, with volumes approaching 4,900–4,930 cars/hour, and lighter during cold months (November–February, ~4,380–4,720). One notable exception: **July 2016** saw a sharp drop to ~3,950 cars/hour, likely due to road construction on Westbound I-94.*

TIME INDICATORS

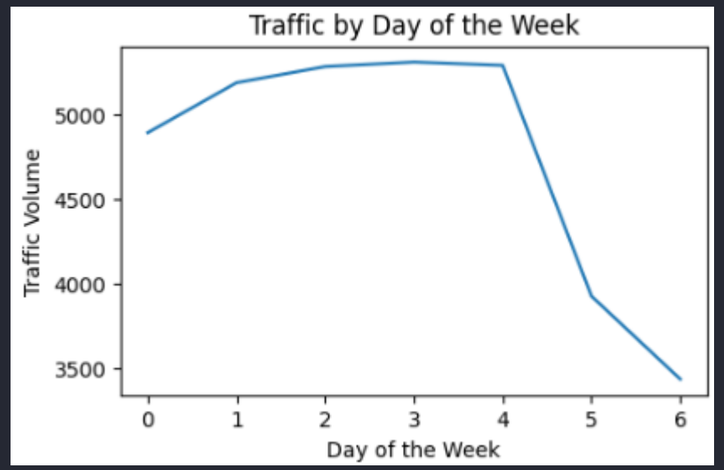
Traffic by Day of Week & Hour

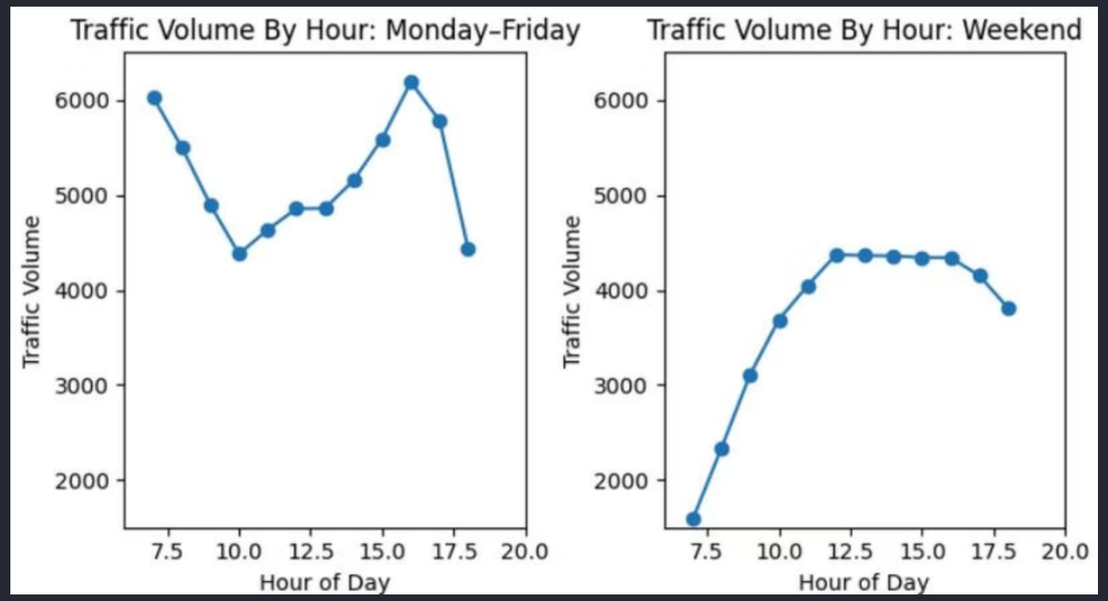
Weekdays (Mon–Fri)

*Traffic regularly exceeds **5,000–5,300** cars/hour, peaking Thursday. Rush hours at **7 AM (~6,000)** and **4 PM (~6,200)** are the busiest periods of the week.*

Weekends (Sat–Sun)

Traffic drops sharply — Saturday averages ~3,900 and Sunday ~3,400 cars/hour. Weekend traffic peaks midday (~4,400) with no sharp rush-hour spikes.





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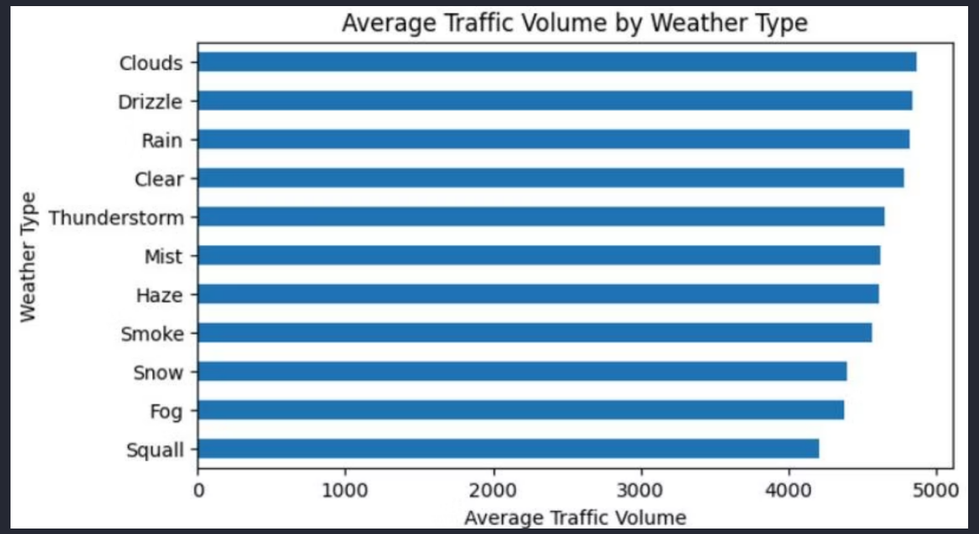
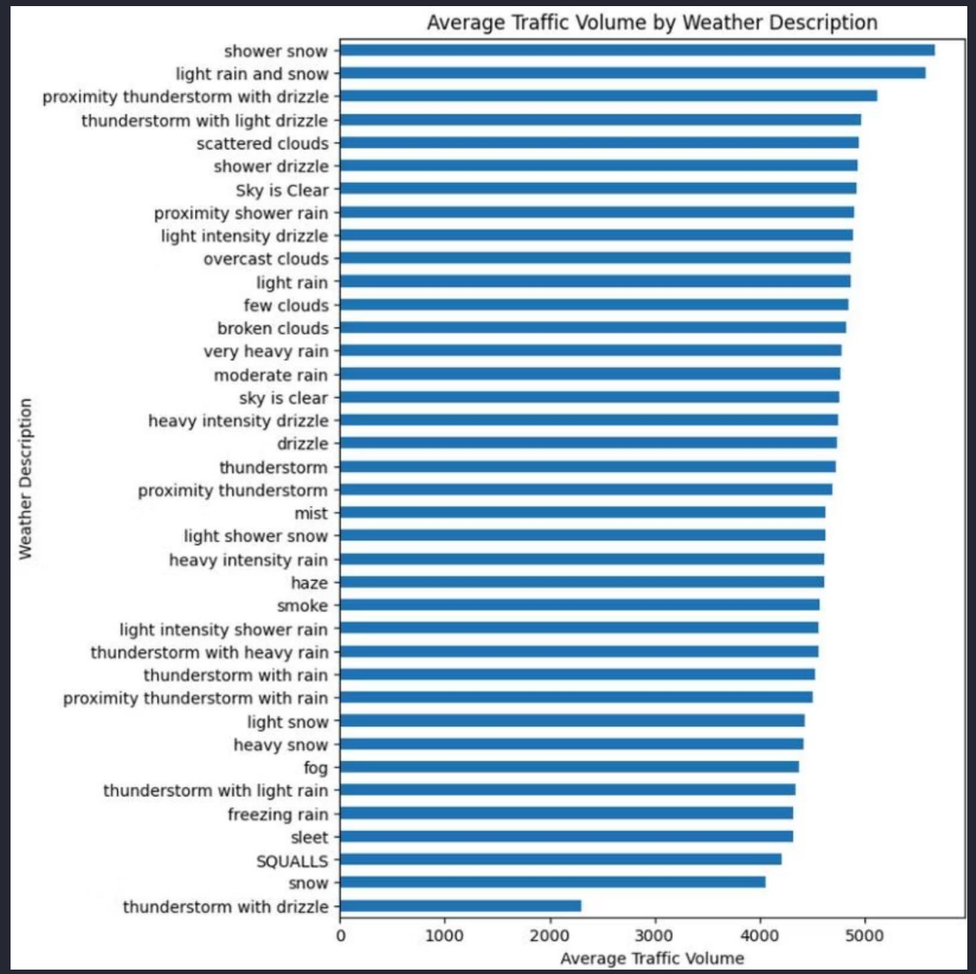
WEATHER INDICATORS

Traffic by Weather Type & Description



*Broad weather categories show little variation — all average below 5,000 cars/hour. Weather type alone is a **weak indicator** of heavy traffic.*



*Detailed descriptions reveal that **"Shower snow," "Light rain and snow,"** and **"Proximity thunderstorm with drizzle"** exceed 5,000 cars/hour — suggesting people drive more in moderately bad weather rather than walking or biking.*

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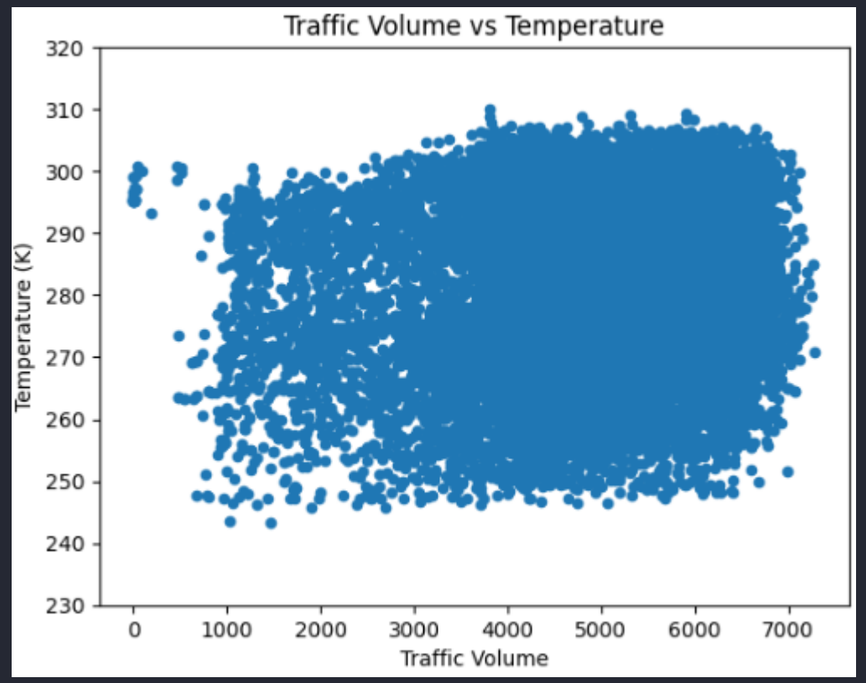

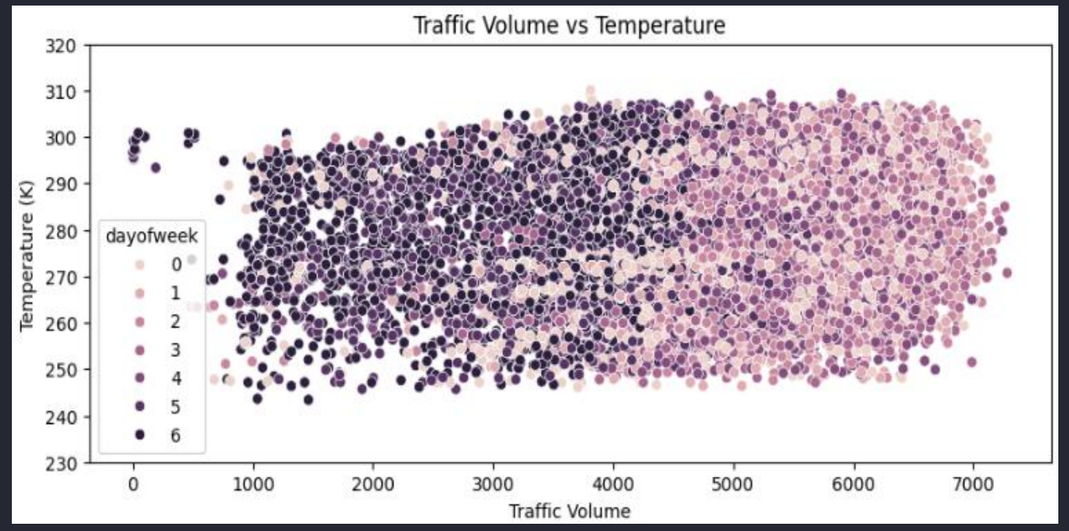

WEATHER INDICATORS

Temperature & Precipitation Correlations

Temperature

*Correlation: **+0.128**. Weakest positive link — not a reliable predictor.*

Rain (1h)

*Correlation: **+0.004**. Nearly zero — rainfall has no linear effect on traffic.*

Snow (1h)

*Correlation: **+0.001**. Essentially no relationship with traffic volume.*

Cloud Cover

*Correlation: **−0.033**. Slight negative — negligible impact overall.*

Temperature shows the strongest but still weak correlation. Rain, snow, and cloud cover have little to no linear relationship with traffic volume.

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WEATHER INDICATORS

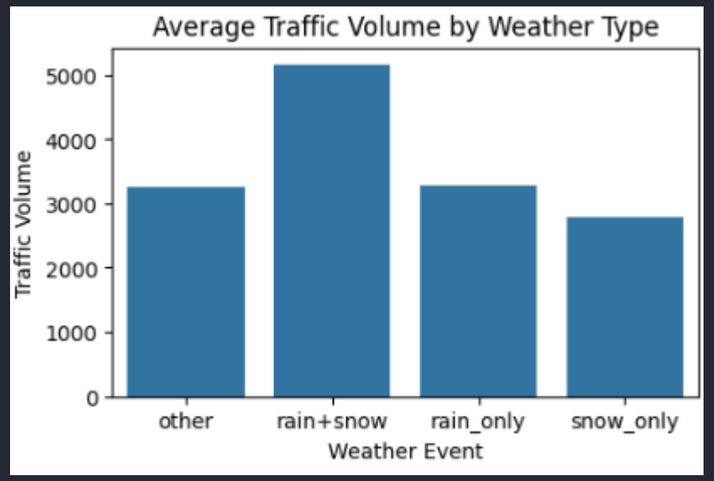
Combined Weather Events & Holiday Effect

Rain + Snow = Highest Traffic

*When rain and snow occur simultaneously, average traffic reaches **5,167 cars/hour** — the highest of any weather combination. Rain only (~3,290) and other conditions (~3,258) are moderate. Snow only produces the **lowest** traffic (~2,788), suggesting snow alone discourages travel.*

Holiday Traffic

*The daytime dataset contains only non-holiday records. On non-holiday days, median traffic is ~**4,800 cars/hour**, IQR from ~4,200 to ~5,500. Very low outliers (0–2,200) likely reflect late-night or disrupted periods. Traffic on non-holidays is **consistently high but variable**.*



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Key Findings & Conclusions

🗓️ Season

*Traffic is heavier during **warm months (March–October)**. July 2016 was an exception due to likely road construction.*

📅 Day of Week

***Weekdays** consistently exceed 5,000 cars/hour. Weekends drop below 4,000 — a clear and strong indicator.*

⏰ Time of Day

*Rush hours at **7 AM and 4 PM** on business days peak above 6,000 cars/hour — the strongest single indicator.*

🌨️ Weather

***Shower snow & light rain+snow** correlate with higher traffic (>5,000). Snow alone reduces traffic. Temperature is a weak predictor.*

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